NEET-SS / FET / INI-SS HIGH-YIELD NOTES

Sports & Arthroscopy

1. ACL and PCL
2. menisci and cartilage
3. patellofemoral instability

ACL and PCL

1. ACL anatomy and biomechanics

- **Origin:** posteromedial aspect of the **lateral femoral condyle** (medial wall of the lateral condyle, intercondylar notch). Landmarks: **lateral intercondylar ridge ("resident's ridge")** marks the anterior (upper in flexion) limit of the footprint; **lateral bifurcate ridge** separates the AM and PL bundles.
- **Insertion:** tibia, anterior to and between the intercondylar eminences, lateral to the medial tibial spine; **anterior horn of the lateral meniscus** blends with it.
- **Length about 32-38 mm, width about 10-11 mm**. **Two bundles (named by tibial insertion):**
  - **Anteromedial (AM): tight in flexion**; primary restraint to anterior tibial translation (anterior drawer at 90 degrees).
  - **Posterolateral (PL): tight in extension**; resists **rotational** loads (pivot shift) and anterior translation near extension.
- **Blood supply: middle genicular artery** (synovial sheath); **nerve:** posterior articular branches of the **tibial nerve** (mechanoreceptors: proprioception).
- **Function: primary restraint to anterior tibial translation (about 85% at 30 and 90 degrees of flexion)**; secondary restraint to internal rotation and varus/valgus.
- **Strength:** about **2,160 N** ultimate tensile load (native).

2. ACL injury

- **Mechanism: non-contact** (about 70%): **deceleration, cutting, pivoting, landing** with the knee near extension, **valgus and external (or internal) rotation**: "**pop**", immediate swelling (**haemarthrosis: ACL tear in about 70% of acute traumatic haemarthroses** in adults).

- **Females 2-8 times higher risk: narrower notch**, **smaller ACL**, **increased Q angle/valgus**, **quadriceps dominance (less hamstring activation)**, **landing in a more upright position with knee valgus**, hormonal (preovulatory phase), **increased posterior tibial slope** (lateral). **Neuromuscular training programmes** (e.g. **FIFA 11+, PEP**) reduce ACL injury risk.
- **Examination:**
  - **Lachman test** (**most sensitive**; knee at **20-30 degrees** of flexion),
  - **Anterior drawer** (90 degrees; less sensitive acutely, hamstring guarding, posterior horn of medial meniscus acts as a "doorstop"),
  - **Pivot shift** (**most specific**; best under anaesthesia; anterior subluxation of the lateral tibial plateau in extension reducing at **20-40 degrees of flexion** via the **iliotibial band** as it moves posterior to the axis of rotation; **requires an intact IT band and MCL**),
  - **KT-1000 arthrometer** (**side-to-side difference over 3 mm** abnormal).
- **Imaging:**
  - **Radiographs: Segond fracture** (**lateral capsular avulsion of the lateral tibial plateau: anterolateral ligament (ALL)/capsule**: **pathognomonic of ACL tear**), **lateral femoral notch sign** (deep sulcus over 1.5-2 mm in the lateral femoral condyle), tibial spine avulsion (children).
  - **MRI:** discontinuity, **bone bruises ("kissing contusions") on the lateral femoral condyle (sulcus terminalis) and posterolateral tibial plateau** (pivot-shift injury), anterior tibial translation over 5-7 mm, uncovered posterior horn of the lateral meniscus, buckled PCL.
- **Associated injuries:**
  - **Meniscal tears: acute: lateral meniscus more commonly torn** (posterior horn, **root tears**); **chronic ACL deficiency: medial**

meniscus (posterior horn) tears increase (it becomes the secondary restraint).

- **Ramp lesions** (**meniscocapsular separation of the posterior horn of the medial meniscus**: probe through the **posteromedial portal/trans-notch view**; often missed on MRI).
- **MCL injuries** (commonest associated ligament), **posterolateral corner (PLC)** (**missed PLC = commonest cause of ACL graft failure from a missed injury**), chondral injuries, **"unhappy triad" (O'Donoghue: ACL + MCL + medial meniscus; actually lateral meniscus more often in acute injuries)**.

3. Management of ACL rupture

- **Non-operative:** "copers" (low-demand, no instability, willing to modify activities): **progressive rehabilitation**. **KANON trial (Frobell): rehabilitation + optional delayed reconstruction** gave similar 2- and 5-year outcomes to early reconstruction in young active adults (**about 50% avoided surgery**). **Increased risk of secondary meniscal tears** with recurrent instability.
- **Operative indications: recurrent instability**, **high-demand pivoting sports**, **repairable meniscal tear**, **combined ligament injuries**, **young age (children: growth-respecting techniques)**, occupational demands.
- **Timing: after resolution of swelling and recovery of full extension and quadriceps control** (avoids **arthrofibrosis**); the "**prehabilitation**" principle. **Early surgery** (within 2-3 weeks) is acceptable when ROM is good; **lock knee from a bucket-handle tear: urgent arthroscopy**.

Graft choice

Graft	Advantages	Disadvantages
Bone-patellar tendon-bone (BPTB)	**Bone-to-bone healing (6-8 weeks), lowest re-rupture in young athletes, strong (about 2,900 N)**	**Anterior knee pain, kneeling pain**, patellar fracture, patellar tendon rupture, **patella baja**, extension loss
Hamstring (semitendinosus ± gracilis, quadrupled)	**Strongest (about 4,000 N quadrupled)**, small incision, **less anterior knee pain**	**Soft-tissue healing (8-12 weeks)**, **tunnel widening**, hamstring weakness (deep flexion), saphenous nerve (infrapatellar branch) injury, **higher re-rupture in young/female athletes with small (under 8 mm) grafts**
Quadriceps tendon (± bone block)	Thick, large cross-section, **low donor morbidity**, increasing use in primary and revision	Quadriceps weakness, patellar fracture (with bone block)
Allograft	No donor morbidity, shorter surgery	**Higher failure in young (under 25), active patients** (especially **irradiated (over 2.5 Mrad)**), slower incorporation, disease transmission (rare), cost

- **Graft diameter: under 8 mm** hamstring grafts have higher revision rates.
- **Graft incorporation: avascular necrosis**, **revascularisation**, **cellular repopulation** and **remodelling (ligamentisation)**: **weakest at about 6-12 weeks**; full maturation 1-2 years.

Technical points

- **Anatomic tunnel placement**: **femoral tunnel low on the lateral wall of the notch (within the native footprint, below the lateral intercondylar ridge)**, drilled via **anteromedial portal**, **outside-in**, or flexible reamers (transtibial drilling often creates a **vertical, non-anatomic graft**: rotational instability with a persistent pivot shift).
- **Commonest technical error: femoral tunnel placed too anteriorly** (graft **tight in flexion**: loss of flexion or graft failure/stretching).
- **Tibial tunnel too anterior**: **roof impingement in extension** (loss of extension, **cyclops lesion** risk); too posterior: vertical graft, PCL impingement.
- **Fixation:** interference screws, suspensory buttons (femoral), staples/screw-post (tibial).
- **Lateral extra-articular tenodesis (LET; modified Lemaire)** or **ALL reconstruction** added in **high-risk patients** (young, high-grade pivot shift, return to pivoting sport, generalised laxity, revision): **STABILITY trial** (hamstring ACLR + LET in young patients: **reduced graft rupture/persistent rotatory laxity**).
- **Double-bundle reconstruction:** no clear clinical advantage over anatomic single-bundle.
- **ACL repair** (proximal avulsion types) is being revisited with augmentation; not standard.

Rehabilitation and return to sport

- **Early full extension, quadriceps activation, closed kinetic chain exercises** (less ACL graft strain than open chain in the early phase); **avoid open-chain quadriceps (knee extension) exercises between 0-45 degrees** early.
- **Return to sport: criteria-based** (strength over 90% limb symmetry index, hop tests, psychological readiness) and **not before 9 months**: each month delay in return to sport until 9 months reduces re-injury by about 51% (Grindem, Delaware-Oslo).

- **Contralateral ACL injury** risk is similar to graft rupture risk in young athletes.

Complications

- **Arthrofibrosis/loss of extension** (commonest complication; **cyclops lesion** = fibroproliferative nodule anterior to the graft: **loss of terminal extension with a "clunk"**: arthroscopic excision).
- **Graft failure: early (technical: tunnel malposition, fixation failure)**, **late (re-injury)**; **missed PLC/medial injuries**; **increased posterior tibial slope (over 12 degrees)**.
- **Infection** (**coagulase-negative staphylococci**, *S. aureus*; **pre-soak graft in vancomycin** reduces infection), **tunnel widening**, DVT, saphenous nerve injury, anterior knee pain, **osteoarthritis** (higher in meniscectomy cases; reconstruction does not prevent OA).
- **Revision ACLR:** assess **tunnel position, widening (over 15 mm: staged bone grafting)**, **posterior tibial slope (over 12 degrees: slope-reducing osteotomy)**, **varus alignment (HTO)**, missed associated laxities.

4. Paediatric ACL injuries

- **Tibial spine (eminence) fractures** (**Meyers and McKeever classification**: I undisplaced, II hinged anterior elevation, III completely displaced (IIIA, IIIB rotated), IV comminuted): **types I and reducible II = closed reduction in extension, cast**; **III/IV and irreducible II = arthroscopic/open fixation (sutures or screws)**; **entrapment of the intermeniscal ligament or anterior horn of medial meniscus** blocks reduction.
- **Intrasubstance ACL tears in skeletally immature:** early reconstruction (to avoid meniscal/chondral damage) using **physeal-sparing (all-epiphyseal, Micheli-Kocher extraphyseal ITB)** or **transphyseal** techniques (with soft-tissue grafts, small vertical central tunnels, no hardware across the physis) depending on

remaining growth (Tanner stage, bone age); risk of **growth disturbance (angular deformity, leg-length discrepancy)**.

5. PCL anatomy and biomechanics

- **Origin: lateral surface of the medial femoral condyle** (anterior notch); **insertion: posterior tibia in a central depression (PCL facet), about 1-1.5 cm below the joint line**.
- **Twice as strong and wider than the ACL** (about 13 mm wide); **two bundles (named by femoral origin): anterolateral (AL, larger, tight in flexion)** and **posteromedial (PM, tight in extension)**.
- **Meniscofemoral ligaments** (from the posterior horn of the lateral meniscus to the medial femoral condyle): **Humphrey (anterior to the PCL)** and **Wrisberg (posterior to the PCL)** (at least one in about 90% of knees).
- **Primary restraint to posterior tibial translation (about 90-95%)**; secondary restraint to external rotation (with PLC).

6. PCL injury

- **Mechanism: dashboard injury** (posteriorly directed force on the proximal tibia with the knee flexed), **fall onto a flexed knee with the foot plantarflexed**, **hyperextension** (combined injuries).
- **Examination:**
  - **Posterior drawer test** (**most sensitive**; 90 degrees of flexion): **grading by tibial step-off** relative to the femoral condyles (**normal about 1 cm anterior step-off**): **grade I: 0-5 mm translation (tibia still anterior to condyles)**, **grade II: 5-10 mm (tibia flush with the condyles)**, **grade III: over 10 mm (tibia posterior to the condyles: suggests combined PCL + PLC/posteromedial injury)**.
  - **Posterior sag (Godfrey) sign**, **quadriceps active test** (contraction reduces posterior subluxation), **reverse pivot shift** (PLC).

    - **Dial test** (external rotation asymmetry over 10 degrees): **at 30 degrees only = isolated PLC**, **at 30 and 90 degrees = PLC + PCL**.
- **Imaging: stress radiographs** (posterior displacement **over 12 mm** side-to-side suggests combined injury), MRI, **avulsion fractures of the tibial PCL insertion**.

Management

- **Isolated grade I-II PCL injuries: non-operative**: **quadriceps-strengthening rehabilitation**, **PCL brace (dynamic anterior drawer force, e.g. Jack brace)**: good functional outcomes; natural history may include **medial compartment and patellofemoral OA** in the long term.
- **Bony avulsions of the PCL tibial insertion: early fixation** (posterior approach (Burks and Schaffer) or arthroscopic) gives excellent results.
- **Grade III, combined injuries (PLC/multiligament), symptomatic chronic instability: reconstruction**: **transtibial** (**"killer turn"** of the graft at the posterior tibia: graft abrasion/attenuation) vs **tibial inlay** (open posterior bone block; avoids the killer turn); **single- (AL bundle)** or **double-bundle**.
- **Risks: popliteal artery injury** during tibial tunnel drilling (**guide-wire penetration; keep the knee flexed at 90 degrees**, use a guard, posteromedial portal to visualise).

7. Posterolateral corner (PLC)

- **Components: lateral (fibular) collateral ligament (LCL)**, **popliteus tendon**, **popliteofibular ligament** (the "big three"), plus lateral capsule, arcuate ligament, biceps femoris, IT band, lateral gastrocnemius.
- **Function:** primary restraint to **varus** and **external rotation** (especially at 30 degrees).

- **Injury:** posterolateral force to the anteromedial tibia, hyperextension/varus; **common peroneal nerve injury in about 15-30%**.
- **Signs: varus thrust gait**, **dial test**, **external rotation recurvatum test**, **posterolateral drawer**, **reverse pivot shift**, **varus stress at 30 degrees (and at 0 degrees: suggests combined cruciate injury)**.
- **Imaging: arcuate sign** (**avulsion of the fibular styloid**: attachments of the popliteofibular, arcuate and fabellofibular ligaments): PLC injury; varus stress radiographs.
- **Treatment: acute repair within about 2-3 weeks** (higher failure of repair alone) or **reconstruction (anatomic, e.g. LaPrade)**: **reconstruction more reliable than repair**; address **varus alignment (HTO first)** in chronic cases; **PLC must be addressed with any cruciate reconstruction** (unrecognised PLC injury causes cruciate graft failure).

8. Multiligament knee injury/knee dislocation

- **Schenck classification: KD I** single cruciate + collateral; **KD II** both cruciates; **KD III M/L** both cruciates + **MCL (M) or PLC (L)**; **KD IV** both cruciates + both collaterals; **KD V** fracture-dislocation; C = arterial injury, N = nerve injury.
- **Vascular injury (popliteal artery)** in about **10-30%** (anterior dislocations: intimal stretch; posterior: transection). **ABI under 0.9: CT angiography/vascular surgery**; **normal pulses do not exclude injury**: serial **ankle-brachial index (ABI)** ± angiography. Ischaemia over 6-8 hours: high amputation rate; **prophylactic fasciotomy** after revascularisation.
- **Peroneal nerve injury** in about 25% (especially **KD III L**).
- **Management:** reduce urgently, assess vascular status, **external fixation** for unstable/vascular/open injuries; **early (within 3 weeks) single-stage or staged ligament repair/reconstruction** generally

gives better results than non-operative treatment; **arthrofibrosis** is common.

Exam pearl: **Lachman most sensitive; pivot shift most specific (needs intact ITB/MCL). Segond fracture = ACL tear. Acute ACL: lateral meniscus tears; chronic: medial meniscus. Commonest technical error: femoral tunnel too anterior (tight in flexion). Tibial tunnel too anterior: roof impingement, cyclops lesion.**

Exam pearl: **AM bundle tight in flexion, PL bundle tight in extension (ACL named by tibial insertion); PCL AL bundle tight in flexion (named by femoral origin). Humphrey anterior, Wrisberg posterior to the PCL.**

Exam pearl: **PCL grade III (tibia behind condyles) = suspect PLC. Dial test: 30 only = PLC; 30 and 90 = PLC + PCL. Arcuate sign = fibular styloid avulsion. Missed PLC causes cruciate graft failure. Knee dislocation: check ABI; under 0.9 = angiography.**

Menisci and cartilage

1. Meniscal anatomy

- **Fibrocartilage**: about **70% water**; dry weight mostly **type I collagen (about 90%)**; proteoglycans, cells (**fibrochondrocytes**).
- **Collagen orientation: circumferential fibres (deep, dominant)** convert axial load into **hoop stresses**; **radial ("tie") fibres** resist longitudinal splitting; superficial random mesh.
- **Medial meniscus: C-shaped**, wider posteriorly, **firmly attached to the capsule and deep MCL**: **less mobile (about 2-5 mm excursion)**: **torn more often** (overall). **Lateral meniscus: more circular (O-shaped)**, covers more of the plateau, **more mobile (about 9-11 mm)**; attached to the **popliteus** (popliteal hiatus: no capsular attachment there) and **meniscofemoral ligaments (Humphrey, Wrisberg)**.
- **Blood supply: medial, lateral and middle genicular arteries** via a **perimeniscal capillary plexus**; only the **peripheral 10-30% (medial) and 10-25% (lateral)** is vascular in adults.
  - **Red-red zone** (peripheral, **under 3 mm from the rim**): good healing.
  - **Red-white zone** (**3-5 mm**): moderate healing.
  - **White-white zone** (**over 5 mm**, inner): avascular, poor healing.
- **Function: load transmission** (**about 50% of load in extension and about 85-90% in flexion through the menisci**; **lateral meniscus bears about 70% of lateral compartment load** vs medial about 50%), **shock absorption**, **joint stability** (**posterior horn of the medial meniscus is a secondary restraint to anterior tibial translation**, especially in ACL-deficient knees), **lubrication and nutrition**, proprioception.
- **Total meniscectomy increases contact stresses by about 2-3 times (up to 235%)**: **Fairbank changes** on radiographs after

meniscectomy: **ridge (osteophyte) formation, flattening of the femoral condyle, joint space narrowing**. **Lateral meniscectomy** has worse outcomes than medial (lateral compartment convex-on-convex geometry).

2. Meniscal tears

- **Patterns:**
  - **Longitudinal (vertical): bucket-handle** (displaced longitudinal: **locked knee**; **double PCL sign** on MRI, **"fragment in the notch" sign**; commoner medially),
  - **Radial** (transects circumferential fibres: **loss of hoop stress function**; **"ghost meniscus"** sign),
  - **Horizontal (cleavage)** (degenerative; **parameniscal cysts**, especially lateral),
  - **Oblique (flap/parrot-beak)**, **complex/degenerative**,
  - **Root tears** (within about 1 cm of the root attachment): **functionally equivalent to total meniscectomy**; **medial posterior root tears in middle-aged women** with a "pop" on squatting (**meniscal extrusion over 3 mm** on MRI); **lateral posterior root tears with ACL injuries**,
  - **Ramp lesions** (posterior medial meniscocapsular junction; ACL-associated),
  - **Meniscal extrusion** (over 3 mm: root tear or degeneration).
- **Clinical:** joint-line tenderness (**most sensitive sign**), **McMurray test**, **Thessaly test (at 20 degrees)**, **Apley grind**; locking, effusion (delayed, hours).
- **MRI: sensitivity about 90-95%** (medial better than lateral); **grade 3 signal (intrasubstance signal reaching the articular surface)** = tear; grades 1-2 are intrasubstance degeneration (not tears).

3. Management of meniscal tears

- **Degenerative tears in middle-aged/older knees** (with or without OA): **physiotherapy first**; **arthroscopic partial meniscectomy (APM) is no better than sham surgery or physiotherapy** for degenerative tears (**FIDELITY**, **ESCAPE**, Katz **MeTeOR**); APM is reserved for **true mechanical symptoms (locking)** or failure of conservative care.
- **Repair indications ("save the meniscus"): peripheral (red-red/red-white), longitudinal, vertical tears 1-4 cm**, **young patient**, **acute**, **stable knee or with concurrent ACL reconstruction** (**healing rates higher with ACL reconstruction: bleeding, growth factors**), **root tears**, **ramp lesions**, **bucket-handle tears** in young patients (reduce and repair).
- **Repair techniques:**
  - **Inside-out** (**gold standard** for mid-body/posterior tears; vertical mattress sutures; **posteromedial approach (protect the saphenous nerve)**; **posterolateral approach: protect the peroneal nerve (retract biceps femoris posteriorly, work anterior to the lateral gastrocnemius)**),
  - **Outside-in** (anterior horn tears),
  - **All-inside** (posterior horn; device-based; **risk to popliteal neurovascular structures in the lateral posterior horn**),
  - **Vertical mattress sutures are biomechanically strongest** (capture circumferential fibres).
  - **Biological augmentation:** marrow venting (notch microfracture), **fibrin clot**, rasping/trephination.
- **Root repair: transtibial pull-out** repair or suture anchor; slows extrusion and OA progression vs partial meniscectomy (in knees without advanced OA).
- **Rehabilitation after repair:** protected weight bearing/limited deep flexion (**avoid squatting beyond 90 degrees for about 3-4**

months).

- **Outcomes: repair success about 70-90%** (better laterally, with ACL reconstruction, acute, peripheral); **failure rate higher medially and in isolated repairs**.
- **APM outcomes:** good for symptomatic unstable flaps in younger knees, but **long-term OA risk** proportional to the amount removed.

4. Discoid meniscus

- **Lateral (almost always)**; **about 3-5% of the population (more in Asians, up to about 15%)**; **bilateral in about 20%**.
- **Watanabe classification:**
  - **Type I: complete** (covers the entire plateau),
  - **Type II: incomplete** (covers under 80%),
  - **Type III: Wrisberg variant** (**no posterior meniscotibial (coronary) attachment**, only the **meniscofemoral ligament of Wrisberg**: **hypermobile**: **"snapping knee syndrome"** in children).
- **Imaging:** radiographs: **widened lateral joint space**, squared lateral condyle, cupping of the lateral plateau; **MRI: three or more contiguous 5-mm sagittal slices showing a continuous "bow-tie"** (meniscal body continuity), or **transverse diameter over 15 mm or over 20% of tibial width**.
- **Treatment: asymptomatic: observe**; **symptomatic: arthroscopic saucerisation (partial meniscectomy to a rim of about 6-8 mm)** ± **repair of peripheral instability (Wrisberg type)**; **avoid total meniscectomy** (early lateral OA, osteochondritis dissecans of the lateral femoral condyle).

5. Meniscal allograft transplantation (MAT) and scaffolds

- **Indication: young (under about 50), symptomatic post-meniscectomy knee** (pain in the meniscus-deficient compartment)

with minimal arthritis (Outerbridge under III), **normal alignment (or corrected by osteotomy)**, **stable knee (or concomitant ACL reconstruction)**, **BMI under about 35**.

- **Contraindications:** advanced OA, uncorrected malalignment/instability, inflammatory arthritis, obesity.
- **Size-matched fresh-frozen allograft** (radiographic sizing: **Pollard method**); fixation with **bone plugs (medial) or a bone bridge/keyhole (lateral: horns close together)**, or soft-tissue sutures.
- **Outcomes: about 70-80% survival at 10 years**; better laterally; **shrinkage, extrusion, tears** common.
- **Collagen (CMI) and polyurethane (Actifit) scaffolds** for partial segmental defects (intact rim needed).

6. Articular cartilage: structure

- **Hyaline cartilage: about 65-80% water**; **type II collagen (about 90-95% of collagen; tensile strength)**; **proteoglycans (aggrecan: GAG chains of chondroitin and keratan sulphate bound to hyaluronic acid by link protein)**: **negative charge attracts water: compressive stiffness (swelling pressure)**; **chondrocytes (about 1-5% of volume)**: produce matrix; **avascular, aneural, alymphatic**: **nutrition by diffusion from synovial fluid**: poor healing.
- **Other collagens: type IX** (cross-links type II), **type X (calcified zone, hypertrophic chondrocytes)**, **type XI** (regulates fibril size), **type VI** (pericellular).
- **Zones:**
  - **Superficial (tangential):** collagen **parallel** to the surface (**highest collagen, resists shear**), **lowest proteoglycan**, flattened cells, **lamina splendens**, **highest water content**,
  - **Middle (transitional):** oblique collagen, round cells,

  - **Deep (radial):** collagen **perpendicular** to the surface, **highest proteoglycan, lowest water**, columns of cells (resists compression),
  - **Tidemark** (separates uncalcified from calcified cartilage),
  - **Calcified cartilage** (type X collagen; anchors to subchondral bone).
- **Lubrication: boundary lubrication (lubricin: proteoglycan 4)**, **fluid-film**, **weeping/elastohydrodynamic**; **biphasic** (solid + fluid) viscoelastic material.
- **Ageing vs OA:**

Parameter	Ageing	Osteoarthritis
Water content	**Decreases**	**Increases (early)**
Collagen	Unchanged/cross-linking increases	Network disrupted
Proteoglycan content	Decreases	**Decreases**
Chondroitin sulphate	**Decreases**	**Increases (chondroitin 4-sulphate)**
Keratan sulphate	**Increases**	**Decreases**
Modulus of elasticity	**Increases (stiffer)**	**Decreases**
Degradative enzymes (**MMP-1, -13, ADAMTS-4/5**, IL-1, TNF)	Normal	**Increased**
Chondrocyte number	Decreases; size increases	Early proliferation (clusters/cloning)

7. Cartilage injury and repair

- **Partial-thickness injuries (not crossing the tidemark): do not heal** (no access to marrow cells).

- **Full-thickness (through the subchondral plate):** marrow cells form **fibrocartilage (type I collagen)**: mechanically inferior to hyaline.
- **Classification: Outerbridge** (grade I softening; **II fissuring/fragmentation under 1.5 cm (0.5 inch) diameter**; **III over 1.5 cm**; **IV exposed subchondral bone**), **ICRS** (0 normal; 1 superficial; 2 **under 50% depth**; 3 **over 50% depth (to calcified layer)**; 4 to subchondral bone).
- **Treatment algorithm for focal chondral defects** (young patients, **correct alignment, stability and meniscal deficiency first**):

Defect	Low-demand	High-demand
Small (under 2 cm^2)	**Debridement/chondroplasty**, **microfracture/marrow stimulation**	**Osteochondral autograft transfer (OATS/mosaicplasty)**, microfracture
Large (over 2-4 cm^2)	Microfracture (less durable), **ACI/MACI**	**Autologous chondrocyte implantation (ACI/MACI)**, **osteochondral allograft (OCA)** (especially with bone loss)

- **Microfracture (Steadman): debride to a stable vertical rim**, **remove the calcified cartilage layer**, awls **3-4 mm apart, 2-4 mm deep**; **fibrocartilage**; **protected weight bearing and CPM for 6-8 weeks**; **results deteriorate after about 2-5 years**, best in **young (under 40), small (under 2-4 cm^2), femoral condyle** lesions; **intralesional osteophytes (subchondral bone overgrowth)** complicate later ACI.
- **OATS (osteochondral autograft): hyaline cartilage** transfer from **less weight-bearing areas** (**lateral trochlea, intercondylar notch**); **defects under about 2 cm^2**; **donor site morbidity** (patellofemoral pain); plugs must be **flush** (**proud plugs: high contact pressures**; recessed: fibrocartilage overgrowth).

- **ACI/MACI: two-stage** (**arthroscopic biopsy of cartilage**, culture, **implantation at about 4-6 weeks** under a **periosteal patch (first generation: hypertrophy commonest complication)**, **collagen membrane (second)** or **matrix-seeded (MACI, third generation)**); **hyaline-like** repair tissue; **larger defects (2-10 cm^2)**; better than microfracture for larger defects in some trials; **slow rehabilitation (about 12 months)**.
- **Osteochondral allograft (fresh, refrigerated, used within about 28 days)**: **large defects with subchondral bone loss**, **OCD**, AVN, salvage; **viable chondrocytes (over 70% viability required)**; **no immune rejection of cartilage (chondrocytes protected in matrix)**, bone portion incorporates via **creeping substitution**.
- **Subchondral insufficiency fracture of the knee (SIFK, formerly SONK): elderly women**, **medial femoral condyle**, often after a **medial meniscus posterior root tear**; **sudden pain**; MRI: **subchondral low-signal line with bone marrow oedema**; protected weight bearing; UKA/TKA if collapse.

8. Osteochondritis dissecans (OCD)

- **Subchondral bone lesion** (repetitive microtrauma, vascular, genetic) with **risk of instability of the overlying cartilage**; **lateral aspect of the medial femoral condyle (about 70%)** (**Wilson sign**: pain at 30 degrees of flexion with internal rotation, relieved by external rotation).
- **Juvenile OCD (open physes)** has a **much better prognosis** (**about 50-75% heal with non-operative care**: activity restriction 3-6 months) than **adult OCD** (closed physes: usually need surgery).
- **MRI signs of instability (De Smet/Kijowski): high T2 signal line behind the fragment**, **cysts** beneath the lesion, **articular fracture line**, **fluid-filled osteochondral defect**.
- **Treatment:**

  - **Stable lesions in juveniles: activity modification, immobilisation**; **retrograde/antegrade drilling** if not healing after 3-6 months.
  - **Unstable but salvageable: fixation** (**bioabsorbable or metal compression screws, chondral darts**, bone pegs).
  - **Unsalvageable/loose bodies: excision + cartilage restoration (OATS, OCA, ACI)**.
- **Other sites: capitellum** (**throwers/gymnasts**: see elbow notes), **talar dome** (**posteromedial (deeper, cup-shaped, atraumatic)** and **anterolateral (shallow, wafer-shaped, traumatic)**).

Exam pearl: **Hoop stresses via circumferential fibres; radial and root tears abolish them (root tear = total meniscectomy). Red-red zone (under 3 mm) heals; repair with ACL reconstruction heals best; vertical mattress strongest. Degenerative tears: physio first (FIDELITY, ESCAPE).**

Exam pearl: **Discoid lateral meniscus: Watanabe III (Wrisberg) is hypermobile: saucerise + repair. MAT: young, meniscectomised, minimal OA, aligned, stable.**

Exam pearl: **Cartilage: type II collagen, aggrecan; superficial zone parallel collagen/highest water; deep zone highest proteoglycan. Ageing: water down, stiffer, keratan up; OA: water UP, keratan down, chondroitin up, MMP-13/ADAMTS up. Microfracture: fibrocartilage (type I); OATS: hyaline, small; ACI/MACI: larger defects; OCA: large osteochondral defects.**

Exam pearl: **OCD: lateral aspect of the medial femoral condyle; juvenile heals non-operatively more often; high T2 line behind the fragment = unstable. SIFK: elderly women, medial femoral condyle, medial root tear.**

Patellofemoral instability

1. Anatomy and biomechanics

- **Patella:** the **largest sesamoid**; **thickest articular cartilage in the body (up to about 5-7 mm)**; **lateral facet larger** than the medial; **odd facet** (medial, contacts in deep flexion).
- **Function: increases the moment arm of the quadriceps (by about 30-50%)**, centralises forces, protects the knee.
- **Contact areas:** at **20 degrees**, the **distal pole** contacts the trochlea; contact moves **proximally** on the patella with increasing flexion; **maximum contact area at about 90 degrees**; **at about 120-135 degrees**, the odd facet contacts the medial femoral condyle.
- **Patellofemoral joint reaction force: about 0.5 x body weight walking**, **3-4 x stairs**, **7-8 x squatting**; increases with flexion (and with open-chain extension near full extension: high quadriceps force).
- **Engagement:** the patella is **laterally positioned in extension** and **engages the trochlear groove at about 20-30 degrees of flexion**: **instability occurs in early flexion (0-30 degrees)**, where the **soft-tissue restraints matter most**.

Restraints to lateral patellar translation

- **Medial patellofemoral ligament (MPFL): primary restraint (about 50-60%) at 0-30 degrees**. **Femoral attachment (Schöttle point): between the adductor tubercle and the medial epicondyle** (just anterior to the posterior cortex extension line, proximal to the posterior origin of the medial condyle on a true lateral radiograph); **patellar attachment: proximal two-thirds of the medial border**. **Injured in almost all (about 90-100%) first-time dislocations**, commonly at the **femoral attachment** (adults) or **patellar attachment** (children).

- **Medial patellotibial ligament, medial patellomeniscal ligament, medial retinaculum** (secondary, more important in deeper flexion), **vastus medialis obliquus (VMO)** (dynamic).
- **Trochlear geometry**: the main restraint **beyond about 30 degrees** of flexion.

2. Risk factors for instability (Dejour "major instability factors")

1. **Trochlear dysplasia** (**most important; about 85-96% of patients with recurrent instability**).
2. **Patella alta** (**delays engagement of the patella in the trochlea**).
3. **Increased tibial tubercle-trochlear groove (TT-TG) distance** (**over 20 mm abnormal**; normal about 10-13 mm; lateralised tubercle).
4. **Patellar tilt** (over 20 degrees on CT).
5. **Other factors: generalised ligamentous laxity (Beighton score)**, **femoral anteversion** (increased), **external tibial torsion**, **genu valgum (increased Q angle)**, **VMO dysplasia**, **female sex**, **young age (adolescents)**, **previous dislocation**, family history.
6. **Q angle:** line from ASIS to the patellar centre and from the patellar centre to the tibial tubercle: **normal about 14 degrees (men) and about 17 degrees (women)**; measured in extension (and in 30 degrees flexion to seat the patella).

3. Imaging and measurements

Radiographs

- **True lateral (posterior condyles superimposed):**
  - **Crossing sign** (**trochlear floor line crosses the anterior border of the lateral condyle**: **trochlear dysplasia**: the trochlea becomes flat at that point),
  - **Supratrochlear spur (bump)**,
  - **Double contour** (hypoplastic medial facet).

  - **Patellar height indices:**
  - **Insall-Salvati ratio** (**patellar tendon length / patellar length**; **normal 0.8-1.2**; **over 1.2 = alta**, under 0.8 = baja),
  - **Caton-Deschamps index** (distance from the **inferior pole of the patellar articular surface to the anterosuperior tibial plateau / articular surface length**; **normal 0.6-1.2**; **over 1.2 = alta**; **not affected by tibial tubercle osteotomy**: preferred for planning distalisation),
  - **Blackburne-Peel ratio** (perpendicular distance from the tibial plateau line to the inferior articular pole / articular length; **over 1.0 = alta**).
- **Axial (Merchant view, 45 degrees knee flexion): sulcus angle (normal about 138 ± 6 degrees; over 145 = dysplasia)**, **congruence angle (normal -6 ± 11 degrees; over +16 abnormal: lateral subluxation)**, **lateral patellofemoral angle** (Laurin: should open laterally).

CT/MRI

- **TT-TG distance** (**over 20 mm = abnormal**; MRI values are smaller than CT by about 2-4 mm), **TT-PCL distance (over 24 mm)** (less affected by trochlear dysplasia/knee rotation).
- **Trochlear dysplasia (Dejour classification):**
  - **Type A: crossing sign**; shallow trochlea (sulcus angle over 145 degrees) but preserved morphology,
  - **Type B: crossing sign + supratrochlear spur**; flat or convex trochlea,
  - **Type C: crossing sign + double contour**; asymmetric facets (convex lateral, **hypoplastic medial facet**),
  - **Type D: crossing sign + spur + double contour**; **"cliff" pattern** (vertical step between facets).
- **Patellar tilt**, **femoral anteversion** and **tibial torsion** (rotational profile CT), **MRI**: **MPFL tear**, **bone bruise** (**medial patellar facet and**

lateral femoral condyle: pathognomonic of transient dislocation), **osteochondral fractures/loose bodies**.

4. Acute first-time patellar dislocation

- **Mechanism: twisting on a flexed, valgus knee with the foot planted** (internal rotation of the femur on a fixed tibia), or direct blow; usually **lateral** dislocation; often reduces spontaneously with knee extension.
- **Examination:** haemarthrosis, **tenderness over the medial retinaculum/MPFL femoral attachment (Bassett sign)**, **apprehension sign (Fairbank)**, **patellar glide (quadrants: over 2-3 quadrants lateral translation = laxity)**.
- **Osteochondral fractures** in about **25-50%** (**inferomedial patellar facet** (commonest) or **lateral femoral condyle**): radiographs/MRI.
- **Management:**
  - **Non-operative for most first-time dislocations** without loose bodies: **brief immobilisation for comfort**, **early ROM, quadriceps (VMO) and hip abductor/external rotator strengthening**, **patellar stabilising brace/taping**.
  - **Surgery after the first dislocation if: osteochondral fracture/loose body** (**fix or remove: arthroscopy**), **substantial MPFL disruption with lateral patellar subluxation on the axial view (relative)**, **high-risk patients** (young with **dysplasia + patella alta + open physes: recurrence risk over 50-70%**).
  - **Recurrence after the first dislocation: about 15-45% overall** (**up to about 70%** in skeletally immature with dysplasia).

5. Recurrent patellar instability: surgical options

- **Principle: address the predominant anatomical abnormalities ("à la carte" surgery, Dejour).**

MPFL reconstruction

- **Workhorse procedure** for recurrent instability: **gracilis or semitendinosus autograft**; femoral fixation at the **anatomic Schöttle point** (**fluoroscopic confirmation**); **graft fixed at about 30-60 degrees of flexion** with the patella centred in the trochlea; **avoid over-tensioning** (**medial patellofemoral overload, pain, loss of flexion, medial instability**).
- **Femoral tunnel malposition: too proximal** (graft **tight in flexion**: **loss of flexion and medial patellar overload**), **too distal** (tight in extension/loose in flexion).
- **Paediatric: femoral attachment is just distal to the distal femoral physis**: drill below the physis or use **physeal-sparing (soft-tissue/adductor sling) techniques**.
- **Complications: patellar fracture** (transverse tunnels across the patella: use **anchors or short sockets**), recurrent instability, stiffness, overconstraint.
- **MPFL repair** has higher recurrence than reconstruction.

Tibial tubercle osteotomy (TTO)

- **Medialisation (Elmslie-Trillat)**: **increased TT-TG (over 20 mm)**: **aim for TT-TG about 10-15 mm**; **avoid over-medialisation** (increases medial tibiofemoral and **medial patellofemoral** contact pressures).
- **Anteromedialisation (Fulkerson)**: **oblique osteotomy**: medialises + **anteriorises (unloads the distal and lateral patellar facets)**: **lateral facet or distal patellar chondral lesions** with instability/malalignment; **contraindicated for proximal/medial facet lesions** (shifts load onto them).
- **Distalisation**: **patella alta (Caton-Deschamps over about 1.2-1.3)**: target about 1.0.
- **Complications: proximal tibial fracture** (**avoid early full weight bearing with Fulkerson: risk of tibial fracture especially at 6-12 weeks**), non-union, **compartment syndrome**, **DVT**, **anterior tibial artery/deep peroneal nerve injury (posterior cortex drilling)**,

painful hardware, **Hauser procedure (posteromedialisation: increases PFJ pressure: abandoned; causes PFJ arthritis)**.

- **Roux-Goldthwait** (lateral half of the patellar tendon transferred medially) in skeletally immature (avoids the tibial tubercle apophysis: **TTO contraindicated with open physes: genu recurvatum**).

Trochleoplasty

- **Severe trochlear dysplasia (Dejour B and D with a supratrochlear spur over 5 mm)**, especially **revision/failed MPFL reconstruction**: **sulcus-deepening trochleoplasty (Dejour/Bereiter)**, **recession wedge trochleoplasty** (for spur). **Complications:** arthrofibrosis, cartilage necrosis, PFJ arthritis; **contraindicated with open physes and established PFJ arthritis**.

Lateral retinacular release

- **Never as an isolated procedure for instability** (no benefit, can cause **iatrogenic medial patellar instability**). **Lateral lengthening** for a **tight lateral retinaculum with a positive patellar tilt test (tilt under 0 degrees: cannot lift lateral facet to neutral)**.

Rotational/angular osteotomies

- **Femoral derotational osteotomy** for **excessive femoral anteversion (over about 25-30 degrees)**; **distal femoral varus osteotomy** for **genu valgum** (over about 5 degrees mechanical valgus) in recurrent instability; **tibial derotation** for external tibial torsion.

6. Special situations

- **Habitual (obligatory) dislocation:** dislocates with **every flexion**, in children: **quadriceps contracture** (**vastus lateralis/iliotibial band contracture**; prior quadriceps injections): **quadricepsplasty/lengthening with realignment**.

- **Congenital (permanent) dislocation:** irreducible from birth; associated with **Down syndrome, nail-patella syndrome (LMX1B: hypoplastic patella, iliac horns, renal disease)**, **arthrogryposis**: early soft-tissue realignment.
- **Medial patellar instability: iatrogenic after lateral release** (vastus lateralis tendon detachment); **lateral retinaculum reconstruction/repair**.
- **Patellofemoral pain syndrome (anterior knee pain): diagnosis of exclusion**; **hip abductor/external rotator and quadriceps strengthening** (best evidence), taping, orthoses; **no surgery**.
- **Patellar tendinopathy (jumper's knee): eccentric/heavy slow resistance exercise**, **decline squats**; **Blazina staging**; surgery (debridement) for refractory cases.
- **Sinding-Larsen-Johansson** (inferior pole traction apophysitis) and **Osgood-Schlatter** (tibial tubercle traction apophysitis): self-limiting; **excise the ossicle** after maturity if symptomatic.

7. Summary: planning algorithm

Pathology	Threshold	Procedure
Torn/insufficient MPFL	Recurrent instability	**MPFL reconstruction (all cases)**
TT-TG increased	**Over 20 mm**	**Medialising TTO** (± anteriorisation for lateral/distal facet wear)
Patella alta	**Caton-Deschamps over 1.2**	**Distalising TTO** (or MPFL alone if mild)
Severe trochlear dysplasia	**Dejour B/D, spur over 5 mm**	**Trochleoplasty** (revision mainly)
Femoral anteversion	**Over about 25-30 degrees**	**Femoral derotation osteotomy**

Pathology	Threshold	Procedure
Valgus	Mechanical valgus over about 5 degrees	**Distal femoral varus osteotomy**
Tight lateral retinaculum	Tilt test negative	**Lateral lengthening** (never isolated release)

Exam pearl: **MPFL = primary restraint 0-30 degrees; femoral attachment (Schöttle point) between the adductor tubercle and medial epicondyle; torn in nearly all dislocations. Dejour major factors: trochlear dysplasia (crossing sign), patella alta, TT-TG over 20 mm, tilt over 20 degrees.**

Exam pearl: **Bone bruise on the medial patellar facet and lateral femoral condyle = transient patellar dislocation. Osteochondral fracture of the inferomedial patella: fix or remove. First-time dislocation without loose body: non-operative.**

Exam pearl: **Proximal femoral tunnel in MPFLR = tight in flexion, medial overload. Fulkerson AMZ unloads lateral/distal facets (avoid for proximal/medial lesions). Caton-Deschamps index is used for distalisation. Never isolated lateral release (iatrogenic medial instability). TTO contraindicated with open physes (recurvatum): Roux-Goldthwait.**